Artificial Neural Networks Assignment 2

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Section: AI-6A

Submitted for: Artificial Neural Networks
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Task 0: Getting Ready with your Dataset

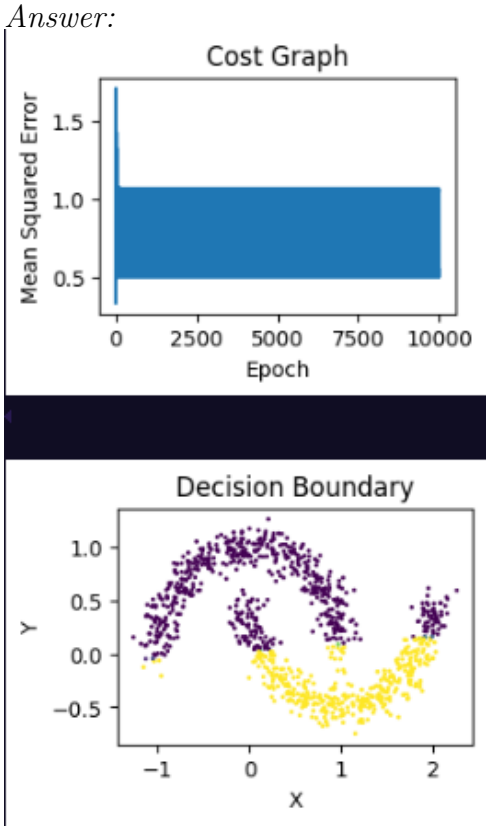
1. **How many features are in the inputs variable, and how much is its length?**
   Answer: The inputs variable has 2 features and a length of 1000.

2. **How many output features are in the outputs variable, and how much is its length?**
   Answer: The outputs variable has 1 feature and a length of 1000

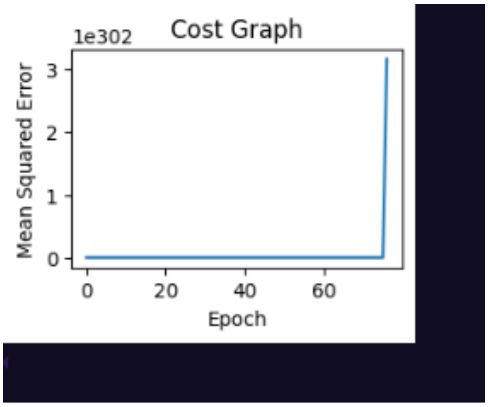
3. **How many points belong to cluster 0, and how many to cluster 1?**
   Answer: Cluster 0 has 500 points, and Cluster 1 has 500 points.

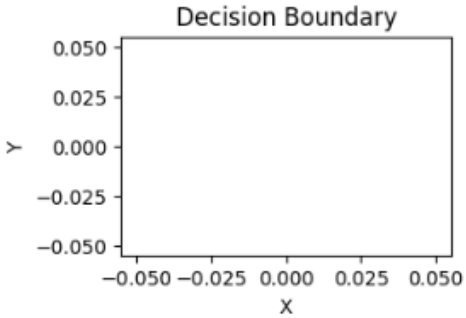
Task 1: 0 Hidden Layer ANN Architecture

1. **Effect of switching from sigmoid to tanh:**
   Answer:

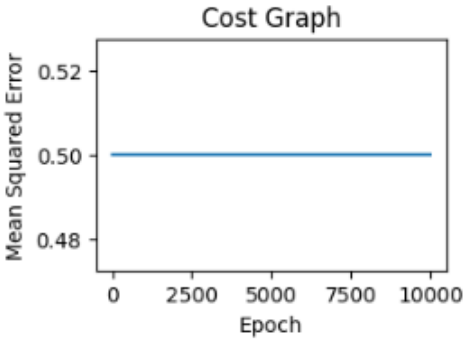


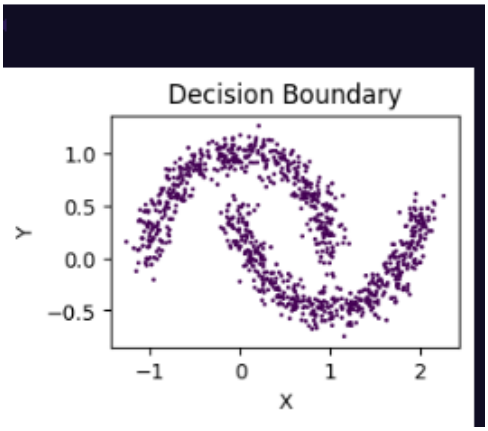
2. **Effect of switching from tanh to ReLU:**
   Answer:





3. **Effect of switching from ReLU to Softmax:**
   Answer:





4. **Learning Rate Experiment:**
   (a) Scatter plots for different learning rates:

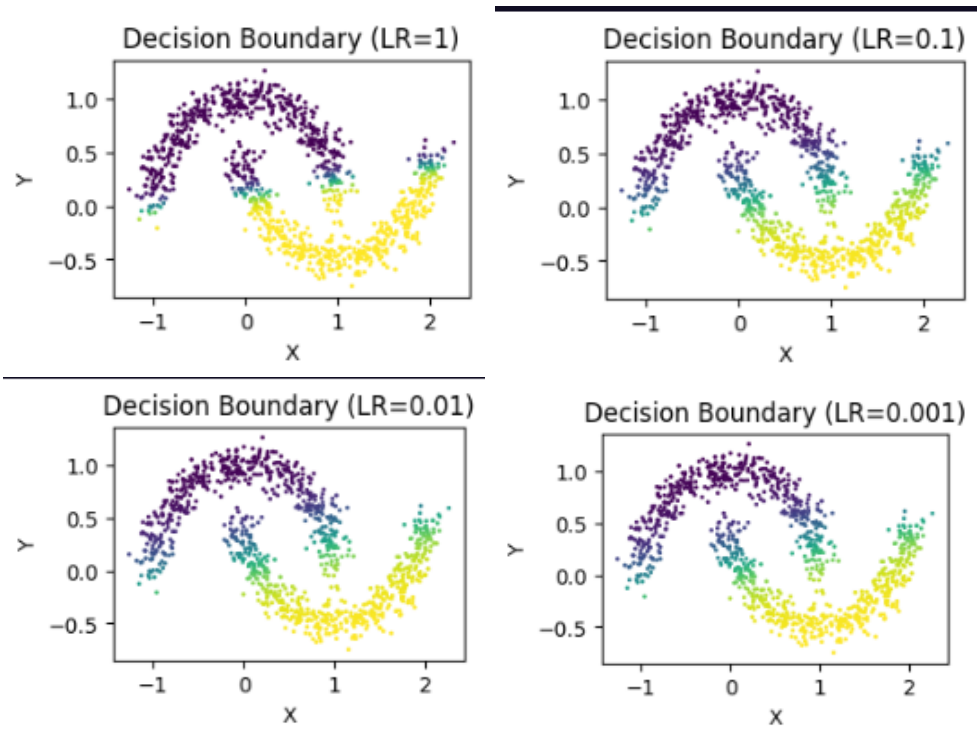


Figure 1: Scatter plots for different learning rates

(b) Minimum error loss:

- lr = 1: 0.09968149695519092
- lr = 0.1: 0.0808948792689298
- lr = 0.01: 0.08089481481670163
- lr = 0.001: 0.08089481481670165

Task 2: 1 Hidden Layer ANN Architecture

1. **Why this shape for weight matrices and biases?**
   Answer: The dimensions of the weight matrices and bias vectors in this neural network architecture are determined by the number of neurons in each layer. The weights input hidden matrix has dimensions (2, 2) because it connects the 2 neurons of the input layer (x and y) to the 2 neurons in the hidden layer. Each input neuron is connected to every hidden neuron. Similarly, the weights hidden output matrix has dimensions (2, 1) as it connects the 2 hidden neurons to the 1 neuron in the output layer. For the bias vectors, bias hidden has dimensions (1, 2) because each of the 2 hidden neurons has an associated bias, and bias output has dimensions (1, 1) since there's a single bias value for the output neuron.

2. **Draw the network diagram:**

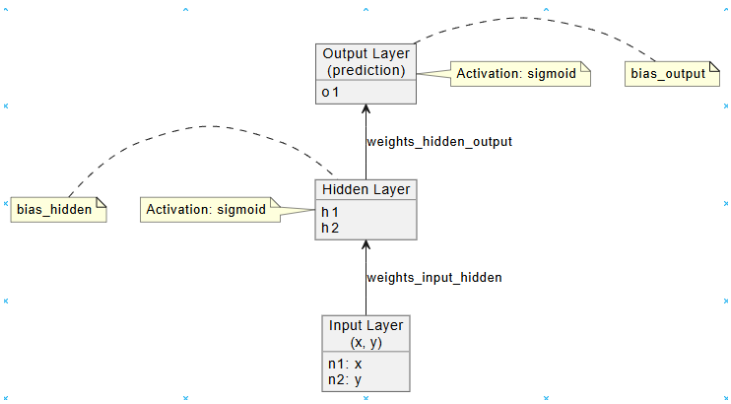


Figure 2: Network diagram for Task 2

3. **Is this result better than Task 1? Why?**
   Answer: Yes it is better and it oscillates less as well because of the added layer.

4. **Effect of larger hidden layer (4 neurons):**
   Answer: 0.00003262218 is less 0.08089481481670163, 4 neurons is better

5. **Network diagram for larger ANN:**

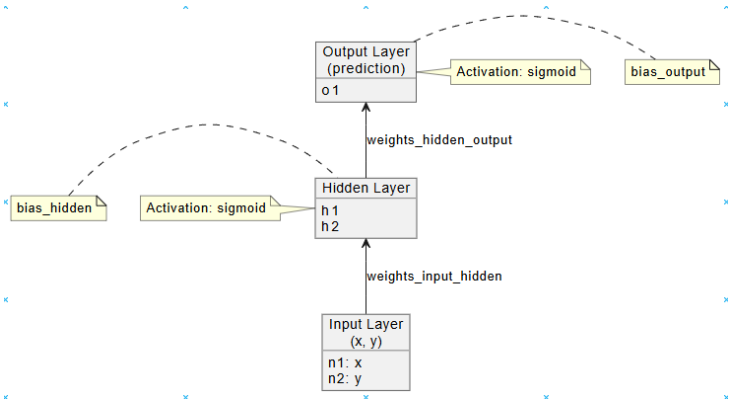


Figure 3: Larger ANN with 4 hidden neurons

6. **Minimum error for each hidden layer size:**

Size	lr=1	lr=0.1	lr=0.01	lr=0.001
2	0.30289	0.07438	0.07059	0.08148
8	0.36770	0.00003	0.30557	0.30557
16	0.07059	0.00033	0.00046	0.00044
20	0.08148	0.08117	0.00625	0.00770
24	0.37571	0.37831	0.00045	0.00509

7. **Minimum error scatter plot: (lr: 0.1, err: 0.00003)**

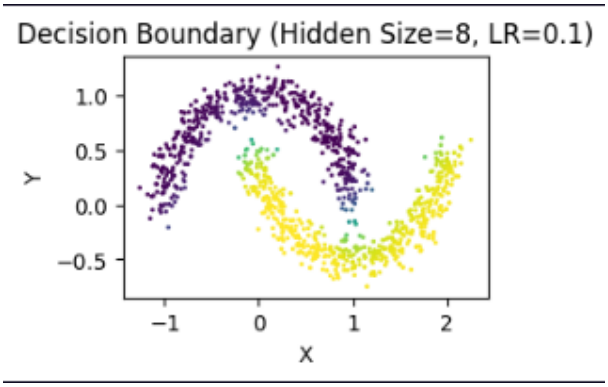


Figure 4: Best performing model

Task 3: 2 Hidden Layers ANN Architecture

1. **Learning rate and hidden layer size that gives better result than Task 2:**
   Answer: — Best Configuration — Learning Rate: 0.5 Hidden Layer 1 Size: 8 Hidden Layer 2 Size: 8 Minimum Error: 0.243664337305907. Unfortunately this was the closest I could get with testing, they weren't upto the benchmark.

Task 4: Change the Dataset (Two Circles)

1. **Best ANN architecture for this dataset:**
   Answer: Given the non-linear, circular nature of the two classes in the "two-circles.csv" dataset, a simple 0-hidden layer ANN (like the one in Task 1) is unlikely to perform well. This is because a 0-hidden layer network can only learn linear decision boundaries.

   A better suited architecture would involve at least one hidden layer (like the one in Task 2) with a non-linear activation function (such as sigmoid, tanh, or ReLU). The hidden layer allows the network to learn complex, non-linear relationships between the input features (x, y) and the output labels.

   For potentially even better performance, especially with the added noise, a 2-hidden layer ANN (like the one in Task 3) might be beneficial. The additional layer could allow the network to learn more intricate features and create a more accurate decision boundary that can handle the noise. The optimal number of neurons in each hidden layer would need to be determined through experimentation (as done in Task 2 and suggested for Task 3).

2. **Can labels be used as z-axis?**
   Answer: If we consider the input features as (x, y) and the label as the z-coordinate, the two classes occupy distinct regions along the z-axis. Therefore, yes, thinking of the labels as the z-axis could make the data linearly separable.

   In this scenario, a simpler model like a linear classifier (which is essentially what a 0-hidden layer ANN learns) could potentially work to separate the data in this 3D space. The linear classifier would aim to find a plane that separates the points with different z-values (labels).

Deliverable

Full notebook is attached along with this report submission.